

FEEDBACK CONTROL SYSTEM

CHAPTER #4

RELATIONSHIP B/W ' ζ ' & 'overshoot'

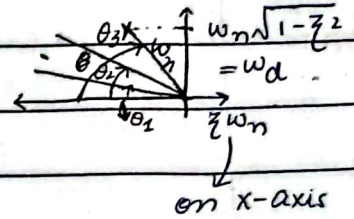
* Every line corresponds to ζ and overshoot.

→ inverse relation (b/w ζ & overshoot)

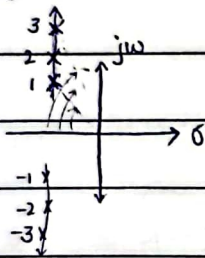
overshoot $\uparrow$ $\zeta \downarrow$

→ direct relation (b/w rise time & ζ)

$\zeta \uparrow$ rise time $\uparrow$



1) .



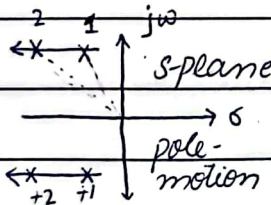
poles on y -axis

(peak time)_{max} = 1 , (peak time)_{min} = 3

(overshoot)_{max} = 3 , (overshoot)_{min} = 1

* all poles on vertical lines have same 'settling time'.

2)



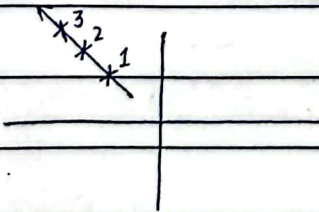
Peak time $\rightarrow$ same

(overshoot)_{max} = 1

(settling time)_{max} = 1

* poles closer to origin have high settling time.

3)



(ζ & overshoot) $\rightarrow$ same

settling & peak time $\rightarrow$ different

(settling time)_{max} = 1

" (peak time)_{max} = 3

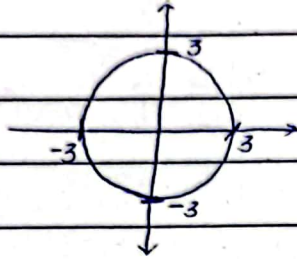
- * Every real poles produces an exponential
- * if pole is ~~neglect~~ far from origin, we neglect it.

Date: _____

Mo Tu We Th Fr Sa Su /

S-PLANE

- circle of 3 in s-plane



Q) * if $\zeta = 0.5$ & $\omega_n = 2$, find transfer function.

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{4}{s^2 + 2s + 4}$$

- if peaks are same - ζ is same & overshoot is same.
- * $\left[\begin{array}{l} \text{smaller } \omega_n \rightarrow \text{less oscillations} \\ \text{high } \omega_n \rightarrow \text{more oscillations} \end{array} \right.$

• $\frac{***}{|} \Rightarrow$ Combination of three first order system $\Rightarrow$ third order system

• $\frac{x}{x \mid x} \Rightarrow$ pole-zero plot of third order system

Date: 27/3

Mo Tu We Th Fr Sa Su

CHAPTER #6 STABILITY

REQUIREMENTS ON THE DESIGN OF CONTROL SYSTEMS:

- Transient response
- Stability
- Steady state error

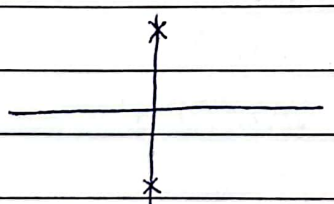
* System should be stable.

- i) Total response (BIBO)
- Bounded input, bounded output $\rightarrow$ stability in terms of total response.
 - Bounded input, unbounded output $\rightarrow$ unstable

$\rightarrow$ Using natural response:

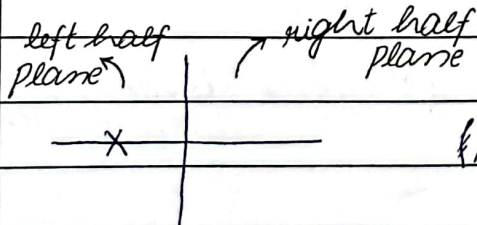
- ii)
- if ' t ' in natural response approaches zero - stable system.
 - if ' t ' approaches ∞ - unstable system.
 - if natural response neither decays nor grows but remains constant / oscillates - marginally stable system.

$\rightarrow$



Oscillatory response
(Marginally stable system)

$\rightarrow$

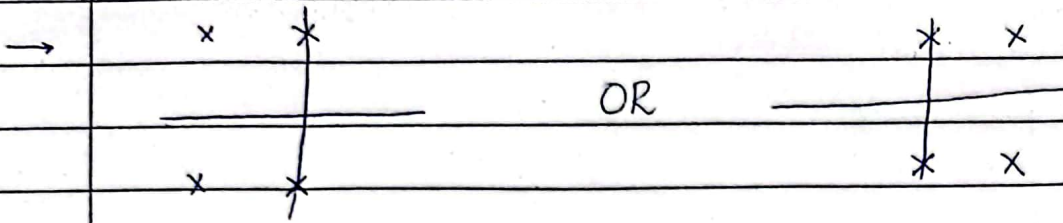


left half plane right half plane

{Pole of multiplicity 'one'

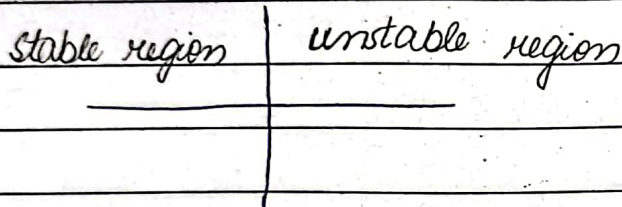
Date: _____

Mo Tu We Th Fr Sa Su /



↳ Poles of multiplicity greater than 'one'
(unstable system)

- right half plane - unstable region
- left half plane - stable region



SUFFICIENT CONDITIONS FOR INSTABILITY

- Signs of denominator's coefficient are not same.

EXAMPLE:

$$1s^3 - 2s^2 + 5s + 3$$

$(+1, -2, +5, +3) \Rightarrow$ signs not same

$$\Rightarrow 1s^3 - 2s^2 + 5s + 3 = 0 \quad (\text{characteristic equation})$$

- Any power of 's' is missing.

EXAMPLE:

$$s^3 - 2s + 3$$

$(s^2 \text{ term is missing})$

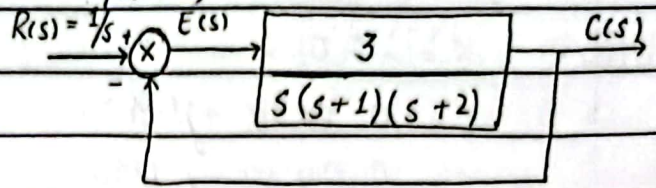
Date: _____

Mo Tu We Th Fr Sa Su

EXAMPLE # 1

(stable system)

$$\frac{3}{s(s+1)(s+2)}$$



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$\Rightarrow 1 + G(s)H(s) = 0$$

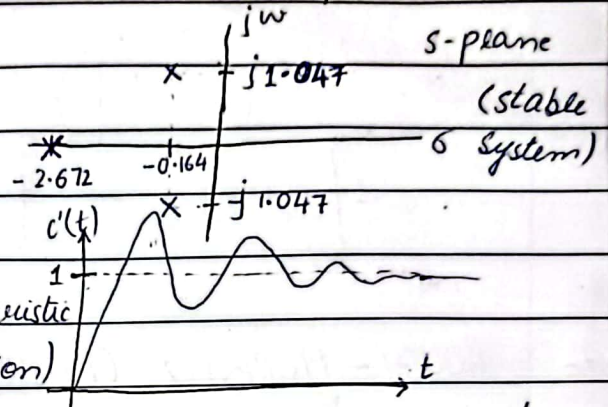
$$1 + \frac{3}{s(s+1)(s+2)} = 0$$

$$\frac{s(s+1)(s+2)}{s(s^2+3s+2)} = 0$$

$$s(s^2+3s+2) = 0 \quad (\text{characteristic equation})$$

$$\Rightarrow \text{poles}$$

Stable system

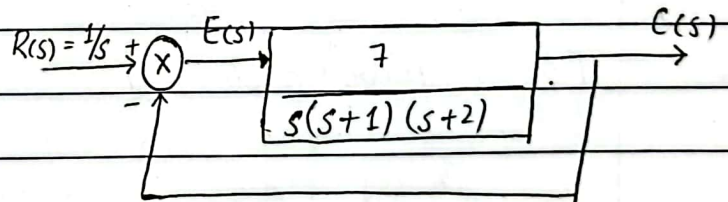


- to check ~~whether~~ whether the system is stable / unstable $\rightarrow$ calculate its poles.

EXAMPLE # 2

(unstable system)

$$\frac{7}{s(s+1)(s+2)}$$



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$1 + \frac{7}{s(s+1)(s+2)} = 0$$

$$\frac{s(s+1)(s+2)}{s(s^2+3s+2)} = 0$$

$$(s^3+3s^2+2s+7) = 0$$

$$s(s^2+3s+2)$$

Unstable system

$$\rightarrow \frac{s(s+1)(s+2) + 3}{s(s+1)(s+2)} = 0$$

$$\Rightarrow s(s^2 + 3s + 2) + 3 = 0$$

$$s^3 + 3s^2 + 2s + 3 = 0$$

factors:

$$s = -2.671$$

$$s = -0.164 + j1.0468$$

$$s = -0.164 - j1.0468$$

Date: _____

Mo Tu We Th Fr Sa Su /

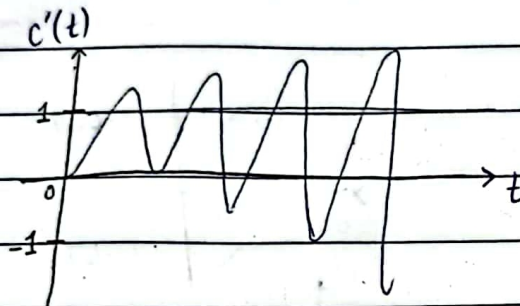
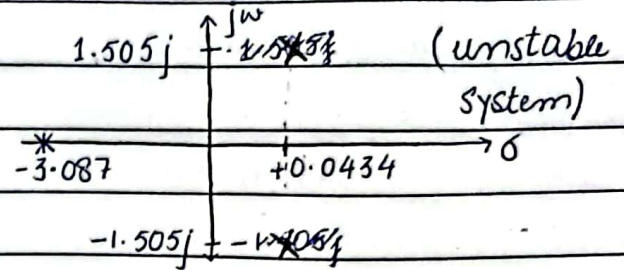
$$s^5 + 3s^4 + 2s + 7 = 0$$

factors:

$$s = -3.087$$

$$s = 0.0434 + j1.505$$

$$s = 0.0434 - j1.505$$



(increase $\Rightarrow$ unstable)

6.2 ROUTH - HURWITZ CRITERION:

→ Used to determine the stability of a system without explicitly solving for its poles.

→ A system is stable if:

- All coefficients of the characteristic equation are positive.
- All determinants must be positive.

For a four order eq: $a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0 s^0 = 0$

s^4	a_4	a_2	a_0 (even 'a's')
s^3	a_3	a_1	0 (odd 'a's')
s^2	$-\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix} = b_1$	$-\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix} = b_2$	$-\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix} = 0$
	a_3	a_3	a_3
s^1	$-\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix} = c_1$	$-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix} = 0$	$-\begin{vmatrix} a_3 & 0 \\ b_3 & 0 \end{vmatrix} = 0$
	b_1	b_1	b_1

AZHAR BOOK

Date: _____

Mo Tu We Th Fr Sa Su

s^0	$- \begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix} = d_1$	$- \begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix} = 0$	$- \begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix} = 0$
	c_1	c_1	c_1

- Denominators ~~equal~~ ⁱⁿ a row are same
- Left column of every determinant in a row are same.

EXAMPLE # 6.1

$$R(s) \rightarrow \frac{1000}{(s+2)(s+3)(s+5)} \rightarrow C(s)$$

$$\Rightarrow R(s) \rightarrow \frac{1000}{s^3 + 10s^2 + 31s + 1030} \rightarrow C(s)$$

s^3	1	31	0
$\div \text{by } 10$ s^2	10 1	1030 103	0
s^1	$- \begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix} = -72$	$- \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} = 0$	$- \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} = 0$
	1	1	1
s^0	$- \begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix} = 103$	$- \begin{vmatrix} 1 & 0 \\ 72 & 0 \end{vmatrix} = 0$	$- \begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix} = 0$
	-72	-72	-72

→ two sign changes in first column. i.e. $1 \rightarrow -72$ (1st change)
 $-72 \rightarrow 103$ (2nd change) $\Rightarrow$ unstable system.

* → The no. of roots of the polynomial that are in right half plane is equal to the no. of sign changes in the first column.

→ Since two poles exist in right half $\Rightarrow$ unstable system

AZHAR BOOK